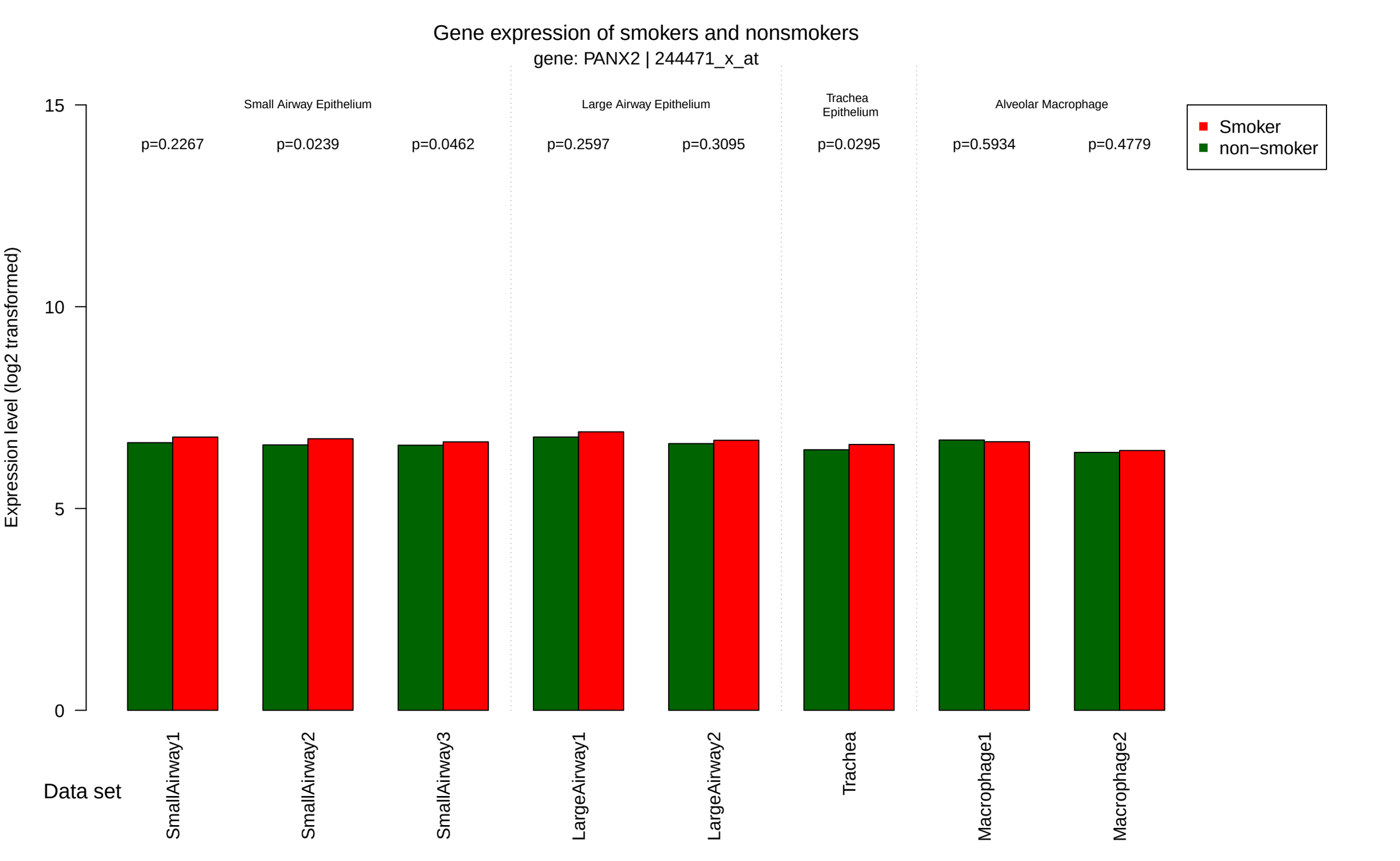
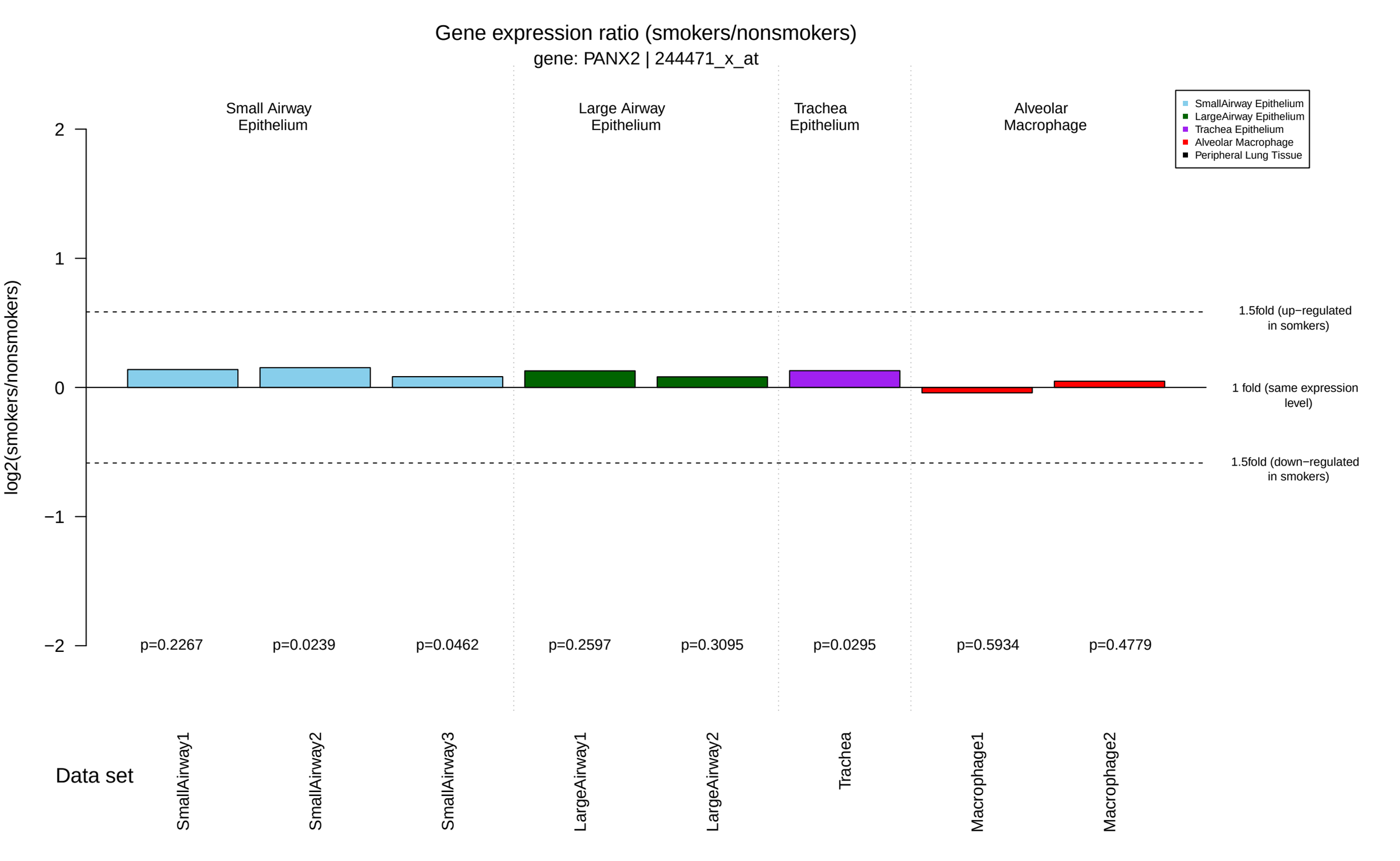
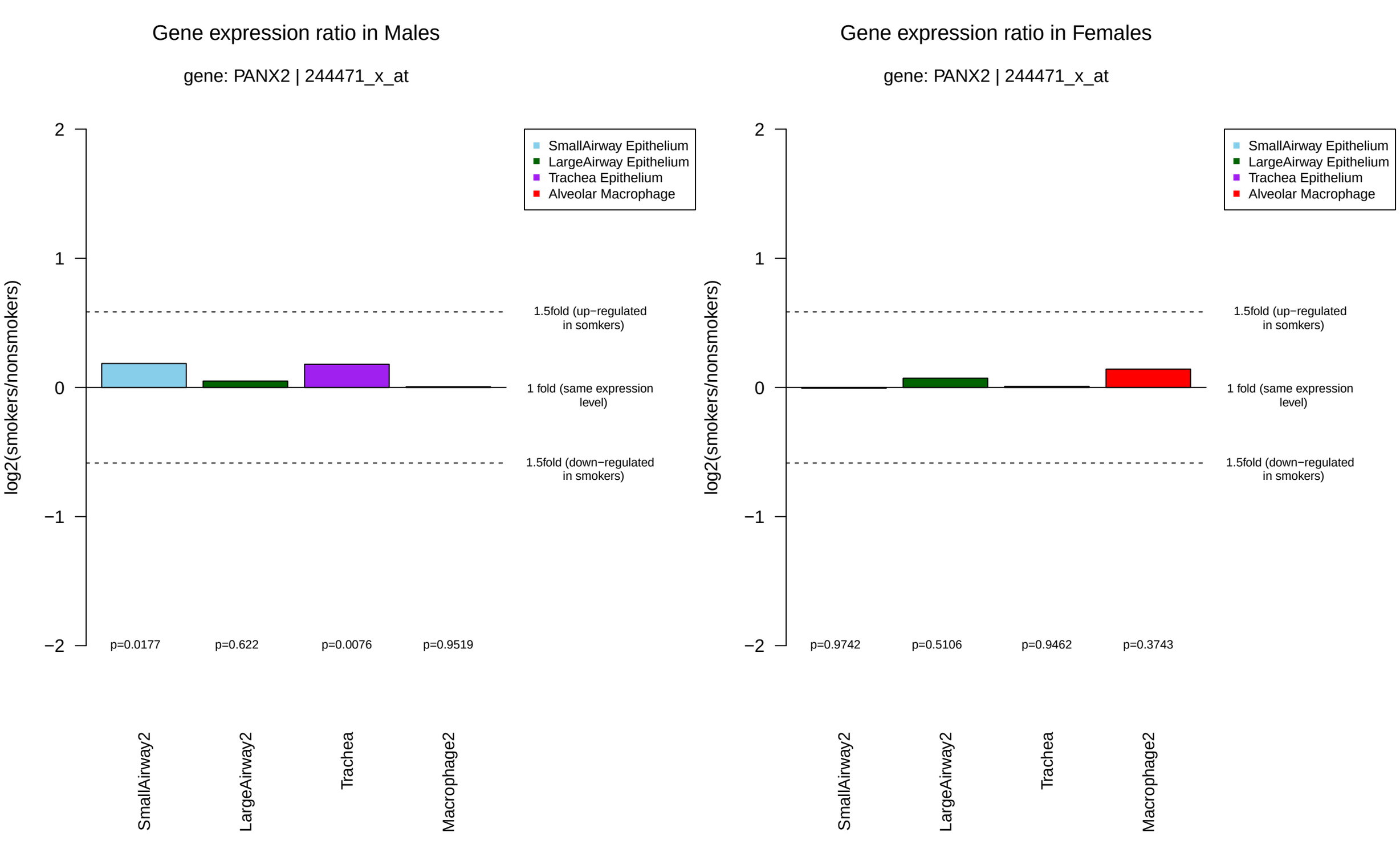
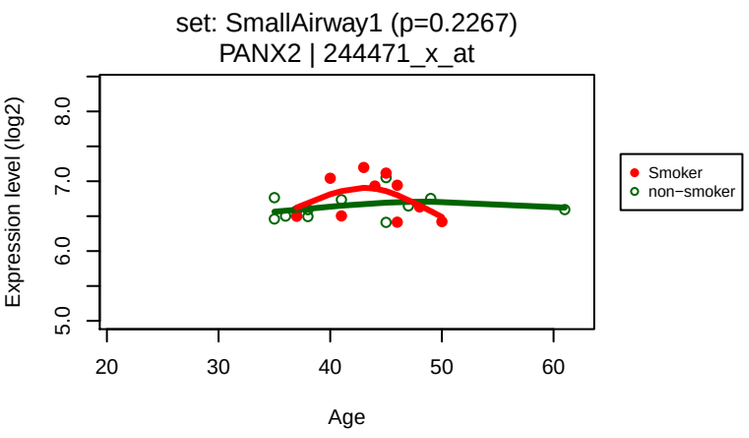
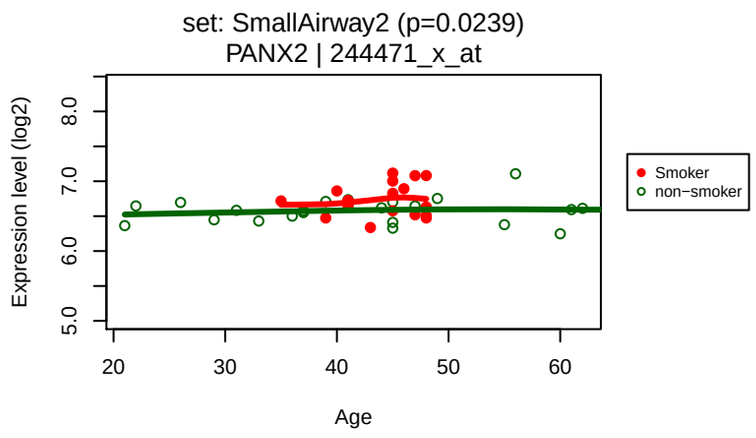
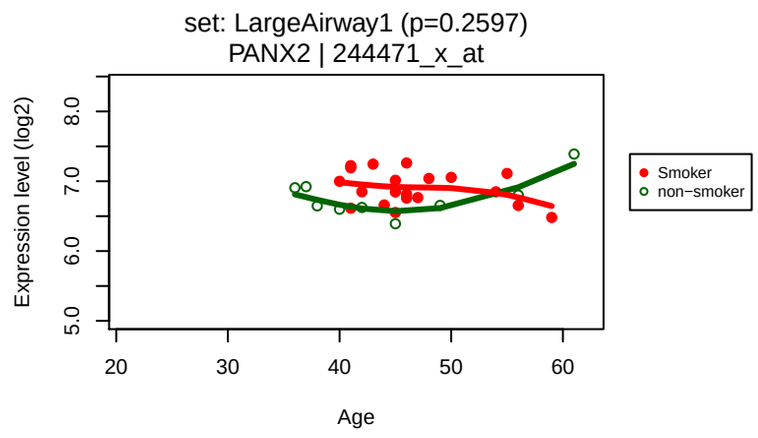

Gene expression of smokers and nonsmokers
gene: PANX2 | 244471_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2267
p=0.0239
p=0.0462
p=0.2597
p=0.3095
p=0.0295
p=0.5934
p=0.4779
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

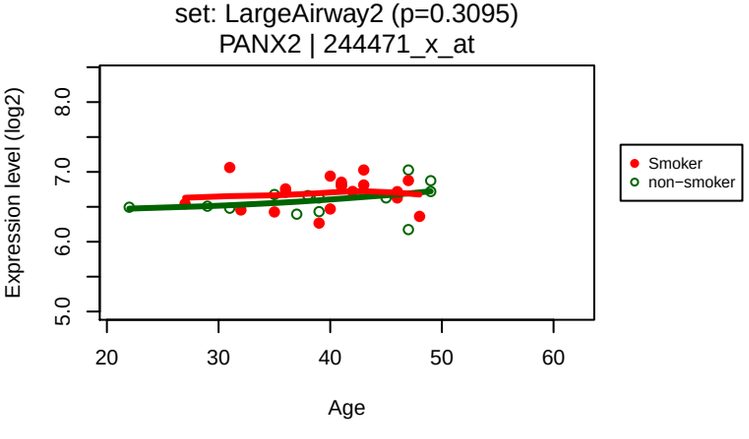

Gene expression ratio (smokers/nonsmokers)
gene: PANX2 | 244471_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2267
p=0.0239
p=0.0462
p=0.2597
p=0.3095
p=0.0295
p=0.5934
p=0.4779
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

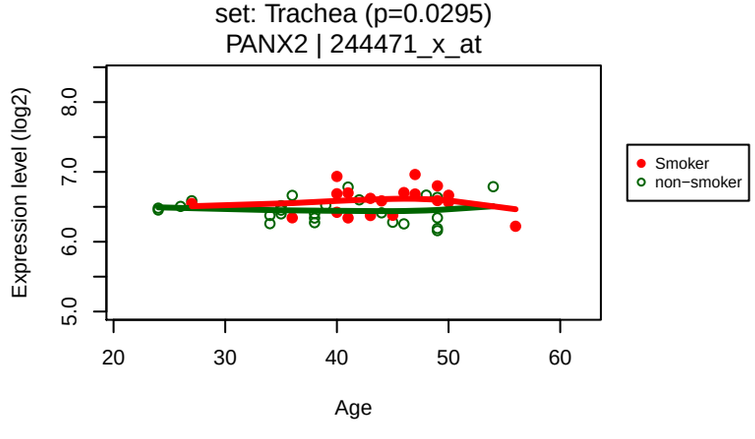

Gene expression ratio in Males
gene: PANX2 | 244471_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0177
p=0.622
p=0.0076
p=0.9519
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PANX2 | 244471_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9742
p=0.5106
p=0.9462
p=0.3743
SmallAirway2
LargeAirway2
Trachea
Macrophage2

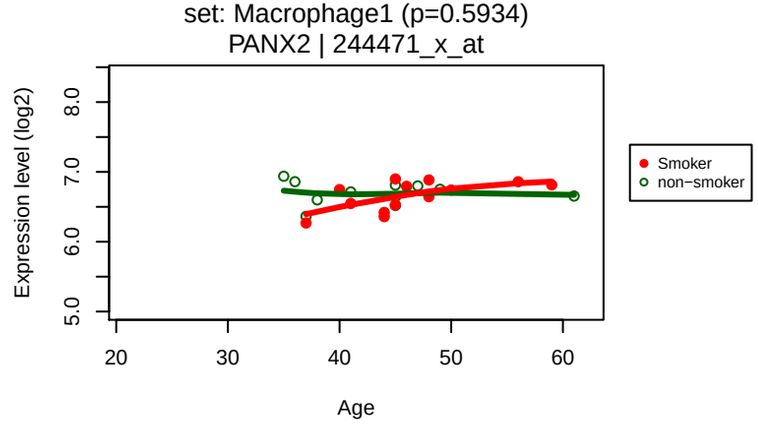

set: SmallAirway1 (p=0.2267)
PANX2 | 244471_x_at
Expression level (log2)
5.0
6.0
7.0
8.0
20
30
40
50
60
Age
Smoker
non−smoker

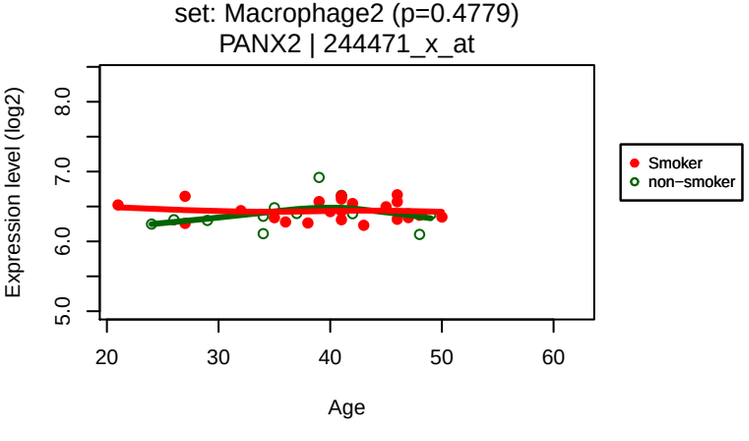

set: SmallAirway2 (p=0.0239)
PANX2 | 244471_x_at
Expression level (log2)
5.0
6.0
7.0
8.0
20
30
40
50
60
Age
Smoker
non−smoker

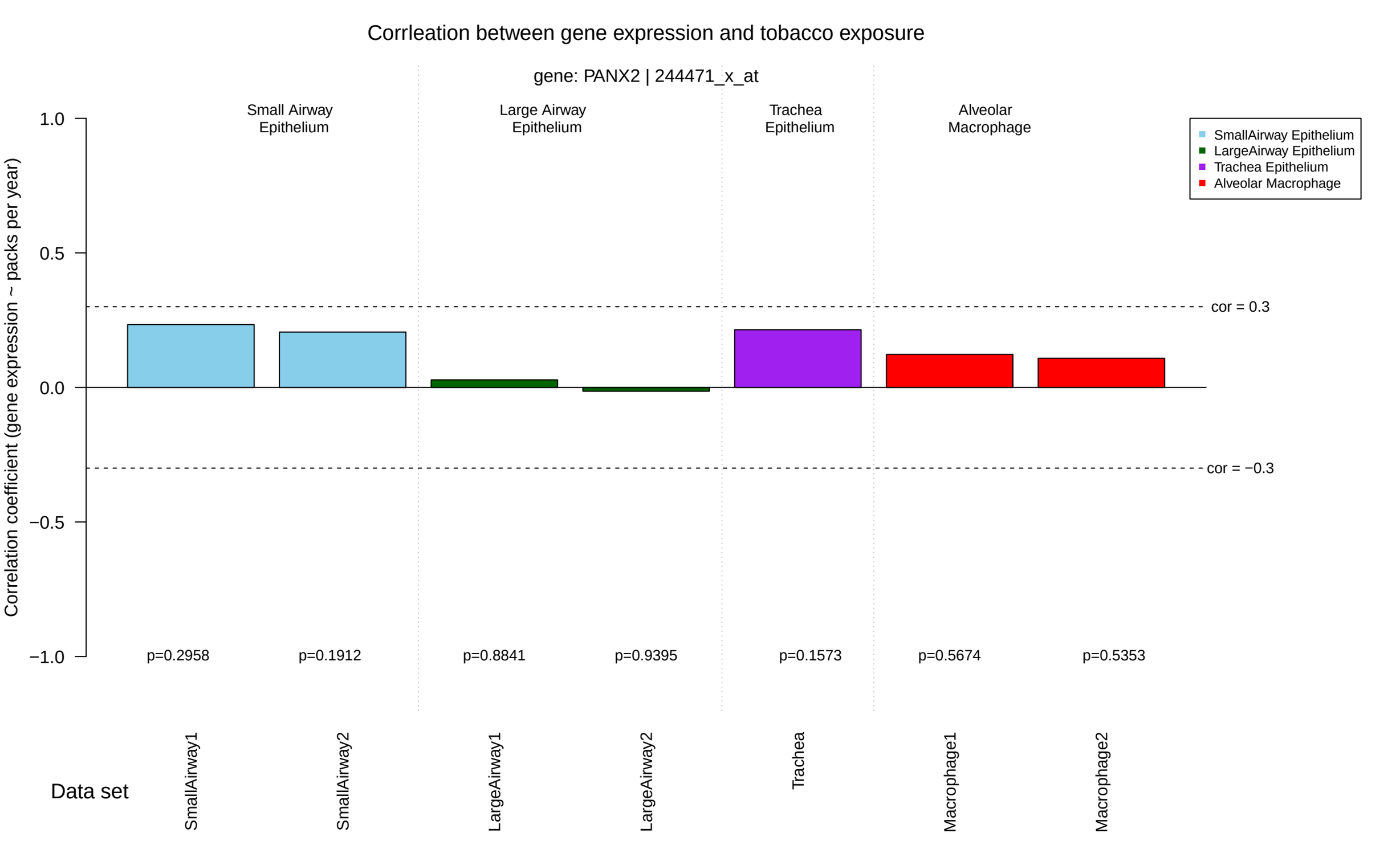

set: LargeAirway1 (p=0.2597)
PANX2 | 244471_x_at
Expression level (log2)
5.0
6.0
7.0
8.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3095)
PANX2 | 244471_x_at
Expression level (log2)
5.0
6.0
7.0
8.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.0295)
PANX2 | 244471_x_at
Expression level (log2)
5.0
6.0
7.0
8.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5934)
PANX2 | 244471_x_at
Expression level (log2)
5.0
6.0
7.0
8.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4779)
PANX2 | 244471_x_at
Expression level (log2)
5.0
6.0
7.0
8.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PANX2 | 244471_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2958
p=0.1912
p=0.8841
p=0.9395
p=0.1573
p=0.5674
p=0.5353
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2